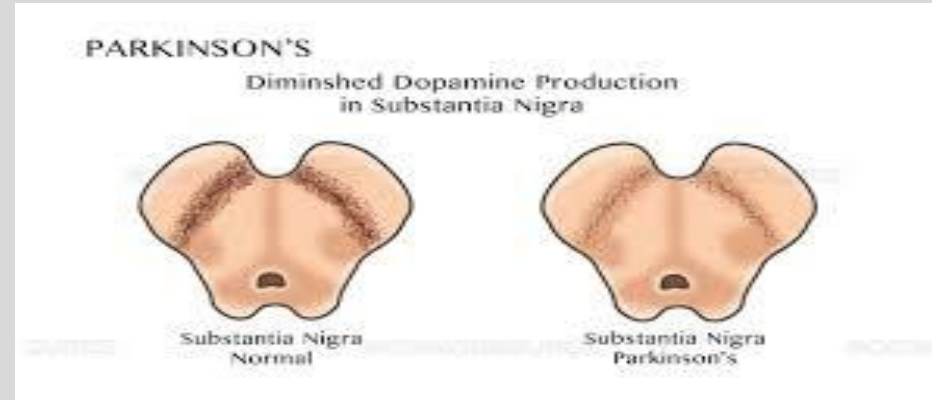


Session Objectives

- Objective 1- **Explain** the motor and non-motor features of PD.
- Objective 2- **Describe** the causes and risk factors of PD.
- Objective 3- **Compare** the symptoms that distinguish idiopathic PD from atypical PD.
- Objective 4- **Utilize** your knowledge to make the diagnosis of PD.
- Objective 5- **Apply** appropriate treatment options and know the adverse effects of PD drugs.

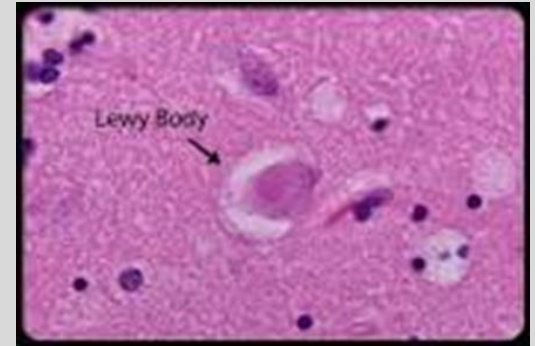
Pathology

- Degeneration of the nigrostriatal **dopaminergic pigmented neurons** in the *ventrolateral region of the **substantia nigra*** is the most consistent neuropathological feature found in all patients with clinical Parkinson's disease.



Pathology

- Misfolded proteins or Lewy bodies are found in the brain.
- Major component protein of **Lewy bodies** is **alpha synuclein**.
- An imbalance between the direct and indirect pathways of the basal ganglia lead to bradykinesia.



Did you know??

- Parkinson's disease symptoms appear when 60% of **dopaminergic neurons are already lost.**
- PD has various clinical presentations, ages of onset, types of non-motor symptoms, and rates of disease progression



How Does it Start?

- Generally vague symptoms:
 - Stiffness (frozen shoulder)
 - Constipation
 - Change in voice
 - **Loss of smell**-anosmia (10 years plus before motor symptoms)
 - REM sleep behavior disorder
 - Repeated stopping of a self winding watch (reduced arm swing)

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Genetics of PD

- 90% do **not** have a family history-therefore sporadic
- PD with GBA (glucocerebrosidase) tend to have a more rapid cognitive decline (highest in Ashkenazi Jews).
- Leucine rich repeat kinase 2 (**LRRK2**) may predict a slower motor progression (Ashkenazi)
- PRKN (Parkin) AR- relatively common when PD onset is in 3rd or 4th decade of life.



Unusual Fact

- Glucocerebrosidase - is a lysosomal enzyme encoded by the GBA1 gene.
- Gene mutation that also causes Gaucher's disease
- Patients with a **GBA1 gene mutation** have a greater chance of developing Parkinson's disease and/or dementia with Lewy bodies due to alpha-synuclein accumulation.



Consider gene testing?

- No family history, but onset before age 50
- Strong family history
- Patients who are either Ashkenazi Jewish, or North African Berber.
- The 7 most common Genes:
- SNCA, PRKN, PINK1, PARK7, LRRK2, VPS35, GBA

Non-motor symptoms

- **Neuropsychiatric:**
 - Depression
 - Anxiety
 - Apathy
 - Impulse control Disorder
 - Psychosis
 - Hallucinations
 - Panic attacks
- **Cognitive:**
 - Executive dysfunction
 - Dementia
- **Dysautonomia:**
 - Orthostatic hypotension
 - Constipation
 - Drooling
 - Urinary incontinence and urgency
 - Sexual dysfunction
 - Altered cardiac reflexes
 - Olfactory dysfunction
 - Gastrointestinal dysfunction
 - Dysphagia
- **Sleep disorders**
 - Restless leg syndrome
 - Insomnia
 - REM sleep disorder
 - Sleep attacks

Neuropsychiatric Symptoms

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- The neurobiology is a complicated mix of PD and other neurodegenerative disease pathology, neurotransmitter deficits, neural circuitry derangement and genetic influences
- **Cognitive impairment, Depression, Anxiety**
- **Psychosis, Impulse Control Disorders, Apathy**



Treatments for Hallucinations

- Quetiapine (Seroquel)- weight gain, tired
- Clozapine (Clozaril)-severe neutropenia
- Pimavanserin (Nuplazid)-prolongs the Q-T interval

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Differential Diagnosis of PD

Atypical parkinsonian syndromes

- Multiple System Atrophy (MSA)
- Progressive Supranuclear Palsy (PSP)
- Corticobasal degeneration (CBD)

Dementia with Lewy Bodies

- Drug induced parkinsonism
- Essential tremor
- Normal pressure hydrocephalus
- Vascular Parkinsonism

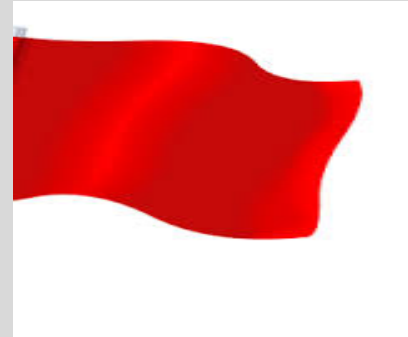


Red Flags for Atypical PD

- Early dementia with hallucinations
- Symmetrical Signs
- Early gait disorder with falls
- Severe orthostasis with syncope
- Alien limb
- Inspiratory Stridor
- Markedly reduced eye movements

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Continuum Lifelong learning In Neurology, Movement Disorders. October 2022 Vol 28, No.

Dementia with Lewy Bodies

- Alpha Synuclein inclusions/ neuronal loss
- REM Sleep behavior disorder
- **Dementia** in the first year of PD onset.
- Visual **hallucinations**- people, animals
- **Fluctuating cognition**
- Parkinsonism in 50%-**symmetric**/action tremor
- **Capgras syndrome/ med sensitivity**

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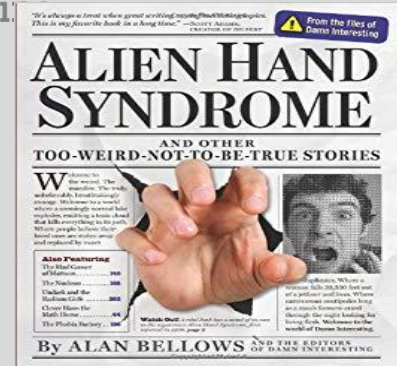


Corticobasal Degeneration (CBD)

- Hyperphosphorylated 4-repeat tau inclusions
- Asymmetric cortical atrophy, usually frontoparietal region
- Markedly asymmetric rigidity
- **Hemidystonia** (muscles contract uncontrollably on one side)
- Variable levodopa response
- Absent resting tremor
- Cortical deficits
 - Apraxia
 - **Alien limb** phenomenon

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Normal Pressure Hydrocephalus (NPH)

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- Classic Triad:
 - Urinary incontinence
 - Dementia
 - Gait disturbance
- **“Wet, wacky, wobbly”**
- Magnetic gait = slow, wide-based gait
- Trial of high volume lumbar puncture (LP) - normal opening pressure
- If LP causes improvement in symptoms, then ventriculoperitoneal (VP) shunt to drain excess CSF



Image: radiopaedia.org

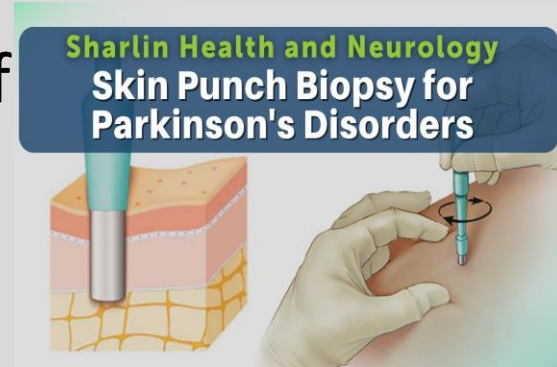
UpToDate – Graff-Radford, Neill R. Normal Pressure Hydrocephalus. 2020.

Alpha Synuclein Testing

- PD, MSA, DLB and pure autonomic failure are all synucleinopathies.
- These conditions have accumulation of phosphorylated alpha synuclein. Aggregates can be found in the central and peripheral nervous systems.
- Tests for biomarkers found in cutaneous nerve fibers and CSF have been developed.

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PET Scan

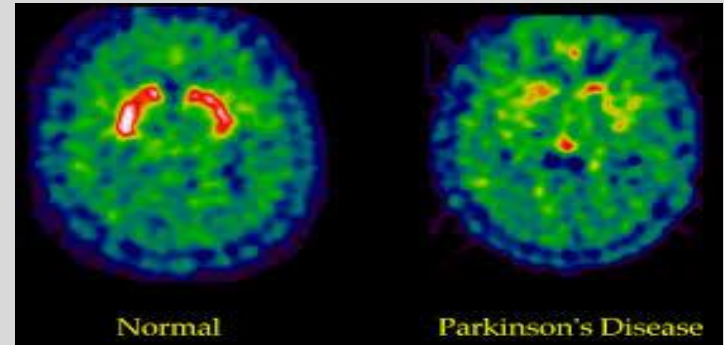
- Reduced uptake of flourodopa in dopaminergic nerve terminals
- Proportional to severity of degeneration in the ipsilateral SN and symptoms in contralateral hemibody.
- Difficult to obtain this test. Not readily available.

Image: atrainctu.com

Content: Bradley's Neurology in Clinical Picture

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Carbidopa/Levodopa

- **Levodopa** is the *gold standard* dopamine replacement therapy for PD
- Given with **Carbidopa** to prevent peripheral breakdown of levodopa and decrease nausea side effect
- Immediate and extended release
- Rytary (IR and ER)
- Crexont (IR and ER)- 10/2024
- Imbrija- inhaled levodopa
- Intestinal gel (Duopa)

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Image:drugs.com

MAO-Inhibitors

- Monoamine oxidase Inhibitors (MAO-B):
 - Selegiline 5mg-10mg QD
 - Rasagiline .5-1mg QD
 - Safinamide (approved 3/21/17) approved as add on treatment for patients already taking carbidopa/levodopa and experiencing off episodes.
- Which one is associated with Serotonin Syndrome?
- Type A or B?
 - Type A breaks down 5HT and NE
 - Type B breaks down DA

Adverse Effects of DAs

- Sleep Attacks
- Ankle/ leg edema
- Impulse control disorder (hypersexuality and gambling)
- Confusion
- Less dyskinesias than levodopa



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Catechol-o-methyltransferase (COMT) Inhibitors

- Entacapone (Comtan)
- Tolcapone (Tasmar) - elevated liver transaminases/ death from hepatic necrosis in 3 pts- not used in clinical practice
- Opicapone (Ongentys) – *new* COMT inhibitor used as add-on treatment to levodopa for off episodes, FDA approved in 2020
- Extend plasma half life of levodopa to reduce wearing off effect
- What is the unusual side effect?



Miscellaneous

- **Trihexyphenidyl** (Artane)-anti-cholinergic treats tremor
- **Benztropine** (Cogentin)-anti-cholinergic treats tremor
- **Amantadine** and extended release (Gocovri) - is an NMDA antagonist and has mild indirect dopaminergic properties. Has anticholinergic properties/ antiglutamatergic activity.
 - Adverse effect = **livedo reticularis**

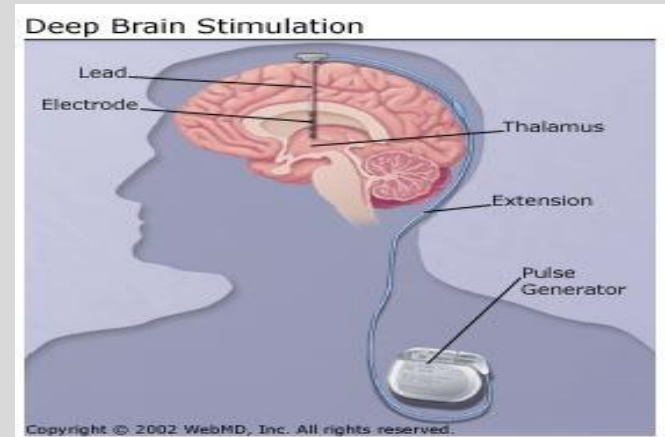


Deep Brain Stimulation (DBS)

- Who is/ is not a candidate?
- Exclusionary criteria
 - Presence of an atypical parkinsonism
 - Unstable psychiatric disease
 - Advanced disease with significant dementia
 - Comorbid medical conditions that preclude surgery

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MRI guided Focused Ultrasound

- Used for PD tremor and dyskinesia
- Unilateral therapy only
- Contraindicated in patients with severe cognitive impairment.
- Patient must be able to tolerate MRI
- Irreversible, unable to adjust postprocedure

What about OMT?

- Trials currently under way...
- Suggests OMT can help improve both motor and non-motor symptoms:
 - Balance, gait and postural stability
 - Treat restricted ROM with muscle energy and myofascial release
 - Autonomic dysfunction (ex. constipation)
 - Address the sympathetics and parasympathetics
 - Stooped posture and restricted breathing
 - Treat with rib raising, thoracic inlet release and lymphatic pumps
 - And so much more...

COVID-19 and PD???

- Three case reports recently published observed acute parkinsonism 2-5 weeks following COVID-19 infection in relatively young patients (ages 35, 45 and 58)
- Studies suggest COVID-19 may cause neurological damage in some patients
- Three proposed mechanisms:
 - Vascular insults to the brain, particularly to the nigrostriatal system
 - Neuroinflammation caused from severe infection → death of dopamine neurons
 - SARS-CoV-2 may be a neurotropic virus, neuroinvasion of the virus → increase in alpha synuclein aggregates
- Also suggests COVID-19 may predispose patients to developing PD later in life...

Méndez-Guerrero A. Acute hypokinetic-rigid syndrome following SARS-CoV-2 infection. *Neurology*. 2020;95:e2109–e2118.

Brundin P, Nath A, Beckham JD. Is COVID-19 a Perfect Storm for Parkinson's Disease? *Trends Neurosci*. 2020 Dec;43(12):931-933.

Rapid Review

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Diagnosis	Key Correlation
Idiopathic PD	“Pill-rolling” resting tremor, rigidity, bradykinesia, postural instability, shuffling gait, stooped posture, masked facies
Lewy body Dementia	Early-onset dementia, hallucinations
PSP	Abnormal eye movements, early-onset gait disorder w/falls
MSA	Autonomic insufficiency, orthostasis, stridor
CBD	Hemidystonia, alien limb, cortical deficits
NPH	Urinary incontinence, dementia, wide-based gait